Measuring the Speed of Sound Underwater

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Introduction

- Objective was to confirm the speed of sound underwater
- Different speed of sound based on the medium that it travels through
- Should be significant;y faster than in air (compressibility, material density).

Methodology

Materials:

- Rangefinder
- Hydrophone
- Audio recording software (audacity)
- Iphone camera (240fps)
- Baking sheet
- Hammer
- Paddleboard

1.

Establish distance between the hydrophone and the mechanical impact and use the rangefinder to record.

2.

Set up slo-mo camera to record both the audacity timer and the mechanical impact.

3.

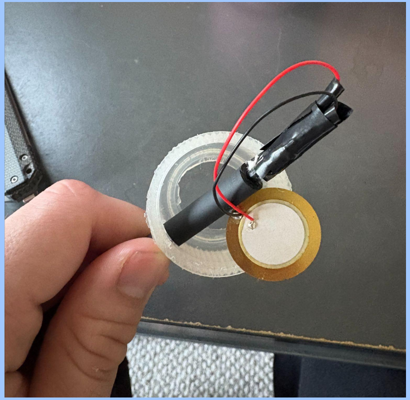
Begin striking. Use the phone video to approximate t = 0 (based on the moment of the hammer strike) on the audacity timeline. Subtract t = 0 from the t of the received waveform from each strike.

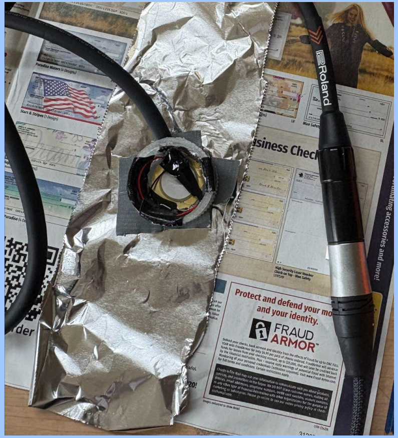
4.

Use the time difference and the measured distance to find the speed of the sound impulse.

Media

The Hydrophone





Audio Recording

tesstdatamp3.mp3

https://drive.google.com/file/d/1KKUO1rEx6GT1s3HjmtV36ztaXLyWUaeE/view?usp=sharing

Formulas Used

Velocity = Distance (m)/Time (s)

$$c(T) \sim 1402.7 + 5T - 0.055T^2 + 0.00029T^3$$

~> Commonly sourced experimentally derived polynomial for the speed of sound underwater in relation to water temperature (C). Water temperature is estimated at approximately 12 degrees C at the time of testing which nets an approximate expected speed of ~1450 m/s. Equation is generally accurate to +- several m/s barring large irregularities in water contents (salt, dissolved sediment, etc.)

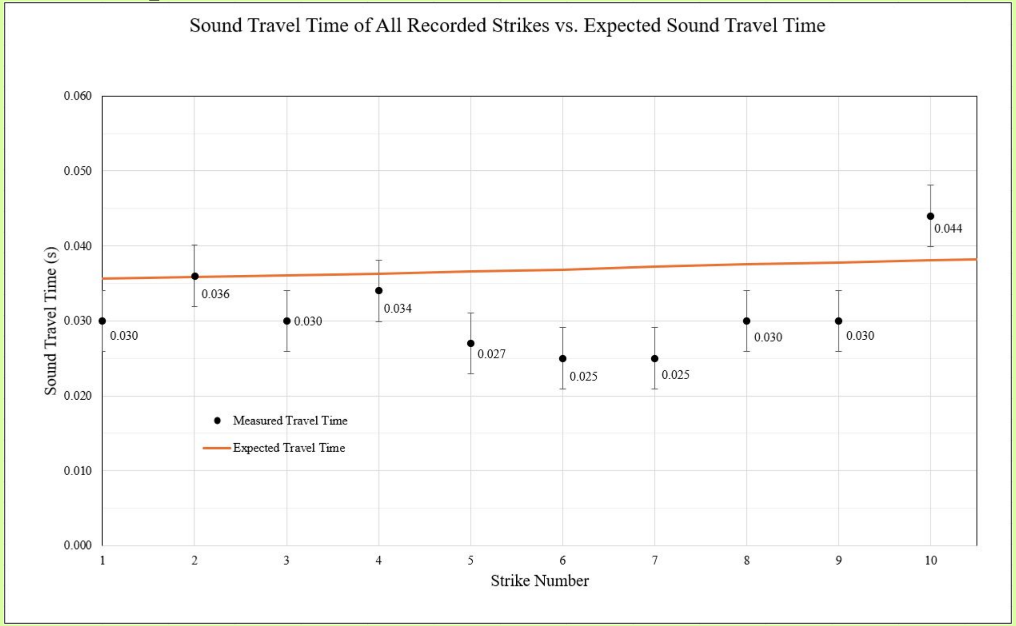
Experimental Data

Strike Number	Time of Strike (s)	Time of First Waveform appearance (s)	Travel Time (s)	Test Distance (m)	Calculated Speed m/s
1	1.887	1.917	0.030	52.8	1760
2	3.117	3.153	0.036	53.1	1475
3	4.484	4.514	0.030	53.4	1780
4	5.878	5.912	0.034	53.8	1582
5	8.863	8.89	0.027	54.2	2007
6	11.915	11.94	0.025	54.6	2184
7	13.588	13.613	0.025	55.1	2204
8	15.323	15.353	0.030	55.6	1853
9	17.009	17.039	0.030	56	1867
10	18.779	18.823	0.044	56.4	1282

Avg Speed (m/s):	Avg T. Time time (s):
1799	0.031
STDEV (+-) speed (m/s)	STDEV (+-) T.Time (s)
295	0.006
95 CI speed (m/s) (+-)	95 CI time (m/s) (+-)(s)
211	0.004

Goal ~ 1450 m/s

Experimental Data



Goal ~ 1450 m/s

Based on our data, we calculated the average speed of sound underwater to be ~ 1800m/s +-211 m/s using a 95 CI.

This is noticeably higher than the theoretical value of approximately 1450m/s, and even higher than many top end speed estimates for more favorable sound travel conditions (more salt, higher temp) which top out around 1580 m/s.

Conclusions and Improvements

Conclusions

1. We could not experimentally corroborate the generally agreed upon speed of sound underwater.
2. The findings do not refute accepted theory, however. Even within the error inherent to our experiment, the measured speed of sound underwater appears to be far higher than the speed of sound in air (343m/s).
3. With improvements to testing methodology there is a high chance of results trending towards theory values.

improvements

1. Faster refresh rates on camera and laptop
2. Better method of making sound
3. Fixed position for both apparatus'
4. Single laptop setup with long cabling and low latency

Thank you